- At critical points function can assume
  - maximum value.
  - minimum value.
  - both maximum and minimum value.
  - **zero value**

- A plane can be perfectly determined by
  - one point.
  - **one point and normal vector.**
  - one point and parallel vector.
  - one normal vector.

- The length and width of a rectangle are measured with the errors of at most 4 % and 3%, respectively. Use differentials to approximate the maximum percentage error in the calculated area.

Solution:

Let
x = length

y = width

A = area

Then $A = xy$ So

$$\frac{\partial A}{\partial x} = y$$

$$\frac{\partial A}{\partial y} = x$$

Therefore,

$$\partial A \approx \frac{\partial A}{\partial x} + \frac{\partial A}{\partial y}\partial y$$

$$= y\partial x + x\partial y$$

We desire percentage change in A, which is relative change multiplied by 100 so let's work out relative change first. This is given by

$$\frac{\partial A}{A} \approx \frac{y\partial x}{A} + \frac{x\partial y}{A}$$

$$= \frac{\partial x}{x} + \frac{\partial y}{y}$$

Since

$$A = xy$$

$$-0.04 \le \frac{\partial x}{x} \le 0.04 \text{ and } -0.03 \le \frac{\partial y}{y} \le 0.04$$

The maximum possible value for $\frac{\partial A}{A}$ given the above constraints. This happens when $\frac{\partial x}{x} = 0.04$ and $\frac{\partial y}{y} = 0.03$, giving $\frac{\partial A}{A} = 0.07$. This is relative error, so the percentage error is 7%.

Convert the Cartesian coordinates $(1, \sqrt{2}, -2)$ into cylindrical coordinates.

Solution:

We are to convert (x,y,z) int (r, θ,z)

Given:

$$x = 1$$

$$y = \sqrt{2}$$

$$z = -2$$

We know that:

$$r = \sqrt{x^2 + y^2}$$

Put value of x and y in the above equation:

$r = \sqrt{(1)^2 + (\sqrt{2})^2}$

$r = \sqrt{1+2}$

$r = \sqrt{3}$

$\tan\theta = \frac{y}{x}$

$\theta = \tan^{-1}(\frac{y}{x})$

$\theta = \tan^{-1}(\frac{\sqrt{2}}{1})$

$\theta = \tan^{-1}\sqrt{2}$

$\theta = 54.73$

$z = -2$

So the cylindrical coordinates are $(\sqrt{3}, 54.73, -2)$

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$\vec{a} = \hat{i} + \hat{j} - \hat{k}$

- A unit vector in the direction of $\vec{a}$ is
  - $\frac{\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right\rangle}{\left\langle \frac{1}{3}, \frac{1}{3}, \frac{-1}{3} \right\rangle}$
  - $\left\langle \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right\rangle$
  - $\left\langle \frac{1}{2}, \frac{1}{2}, \frac{-1}{2} \right\rangle$

Find equation of the tangent plane to the surface $z = x^3 + y^3$ at the point $(2,1,3)$.

Solution:

$f_x = 3x - y$

$f_y = 3y - x$

$f_z = -1$

$f_x(P) = 6 - 1 = 5$

$f_y(P) = 3 - 2 = 1$

$f_z(P) = -1$

Equation of tangent line to the surface through P is:

$5(x-2)+1(y-1)-1(z-3)=0$

$5x-10+y-1-z+3=0$

Hence:

$5x+y-z-8=0$

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Dot product of $\vec{a} = \hat{i}+2\hat{j}+\hat{k}$ *and* $\vec{b} = 2\hat{i}+3\hat{j}+2\hat{k}$ is

- 8
- 5
- **10**
- 13

- Evaluate the double integral $\iint_R x^2 dA$; R is the region bounded by $y = \frac{16}{x}$, $y = x$, *and* $x = 8$.

Solution:

Area of R = $\int \int_R x^2 dA$

$= \int_2^8 \int_{\frac{16}{x}}^{x} x^2 dy dx$

$= \int_2^8 \int_{\frac{16}{x}}^{x} x^2 \Big|_{\frac{16}{x}}^{x} dx$

$= \int_2^8 x^3 - 16x dx$

$$= \int_2^8 \frac{x^4}{4} - \frac{16x^2}{2}$$

$$= \int_2^8 \frac{x^4}{4} - 8x^2$$

$$= \left(\frac{8^4}{4} - 8\times 8\right) - \left(\frac{2^4}{4} - 8\times 2\right)$$

$$= \mathbf{972}$$

- Domain of $\sqrt{x} + \sqrt{y}$ is
  - $x + y > 0$
  - $x \geq 0\ , y \geq 0$
  - $x > 0, y > 0$
  - $x + y \geq 0$

- Use double integrals to compute the volume of the solid bounded by the *cylinder* $x^2 + y^2 = 9$ *and the planes* $y + z = 9$ *and* $z = 0$

Solution:

$$\iint_R (9 - y)da$$

R as a type 1 region we find that:

$$V = \int_{-2}^{2} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (9 - y)\, dydx$$

$$= \int_{-2}^{2} \left[9y - \frac{1}{2}y^2 \right]_{y=-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx$$

$$= \int_{-2}^{2} 18\sqrt{9 - x^2}\, dx$$

$$= 18\left(\frac{1}{2}\pi 9^2\right) = 729\pi$$